InterSen: Boosting LoRa Sensing Capability Under Communication Interference

Qiling Xu[1], Binbin Xie[2], Xiong Jie[3], Lu Wang[4], Zhimeng Yin[1] *
[1]City University of Hong Kong, [2]University of Texas at Arlington, [3]Nanyang Technological University, [4]Shenzhen University

ABSTRACT

LoRa technology holds great promise for wide-area wireless sensing, owing to its long-range connectivity and strong penetration capability. However, existing LoRa sensing systems face a fundamental limitation, i.e., they assume that the gateway only receives signals from the sensing node, without any interference from other communication nodes. This is because transmissions from communication nodes inevitably distort the sensing pattern extracted from the received sensing signal, leading to sensing failure. To address this challenging issue, we propose *InterSen*, a novel solution specifically designed to address communication interference on LoRa sensing. We conduct an in-depth analysis of how communication interference disrupts LoRa sensing and design innovative signal processing methods to recover the sensing pattern corrupted by interference. We evaluate the performance of *InterSen* across three real-world sensing applications, i.e., respiration monitoring, walking sensing, and gesture recognition. Comprehensive experiments demonstrate that *InterSen* achieves accurate sensing even in the presence of multiple communication nodes. This brings LoRa sensing one step closer to practical deployment in real-world scenarios.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**.

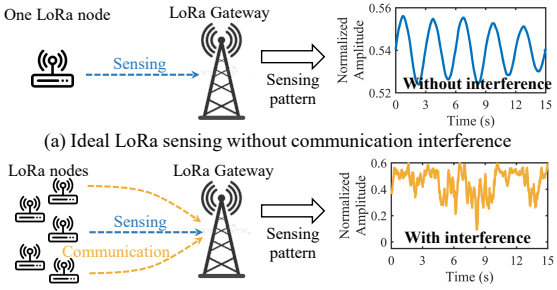
KEYWORDS

LoRa sensing, Contact-free sensing, Communication interference

ACM Reference Format:
Qiling Xu[1], Binbin Xie[2], Xiong Jie[3], Lu Wang[4], Zhimeng Yin[1] . 2025. InterSen: Boosting LoRa Sensing Capability Under Communication Interference. In *International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc '25), October 27–30, 2025, Houston, TX, USA.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3704413.3764417

1 INTRODUCTION

Wireless sensing has demonstrated great potential by utilizing radio frequency (RF) signals to sense human motion without direct physical contact. Various types of wireless signals, such as WiFi [24], mmWave [1], RFID [18], UWB [23], and acoustic signals [21], have been explored for this purpose. A key challenge common to these contactless sensing approaches is the limited sensing range, stemming from their reliance on reflected signals from the target, which are typically much weaker than direct path signal. Recently, LoRa-based sensing has emerged as a promising solution to the short sensing range limitation owing to its chirp signal design and lower frequency. For instance, the state-of-the-art WiFi sensing range is typically limited to a few meters, making it difficult to cover the whole apartment. In contrast, LoRa sensing can achieve a range of hundreds of meters (e.g., 138 m [27]). This makes LoRa-based sensing very suitable for applications that require large-area coverage.



(a) Ideal LoRa sensing without communication interference

(b) Practical LoRa sensing with communication interference

Figure 1: Contact-free LoRa sensing in existing LoRa networks: ongoing communication transmissions from other LoRa nodes can disrupt sensing.

Although LoRa-based contact-free sensing has achieved promising progress [27, 30], one practical issue still hinders the integration of LoRa sensing into existing LoRa infrastructure: current LoRa sensing techniques commonly assume that the received signal is free of communication interference, neglecting the presence of co-existing communication nodes. In practice, however, sensing signals can be disrupted by interference from communication signals—an issue that becomes increasingly severe with the rapid expansion of LoRa networks. To investigate the impact of LoRa communication on LoRa sensing, we conduct preliminary experiments and our findings reveal that the presence of LoRa communication severely degrades LoRa sensing performance. As shown in Fig. 1(a), without any ongoing LoRa communication, a clear respiration pattern can be extracted at the gateway, and the calculated respiration estimation error is 0.3 bpm (breaths per minute). However, when LoRa communication packets are transmitted concurrently in Fig. 1(b), the respiration pattern becomes severely distorted, increasing the error to 4.1 bpm (far exceeding the acceptable threshold of 1 bpm).

Since the interference issue caused by multiple concurrent LoRa transmissions has been extensively studied in LoRa communication [4, 15, 25, 32], a natural question arises: *can these communication techniques be directly adopted to mitigate communication interference on sensing?* To answer this question, we conduct in-depth analysis of how communication affects LoRa sensing. Our findings surprisingly reveal that *communication interference affects sensing*

*Corresponding author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
MobiHoc '25, October 27–30, 2025, Houston, TX, USA
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1353-8/25/10…$15.00
https://doi.org/10.1145/3704413.3764417

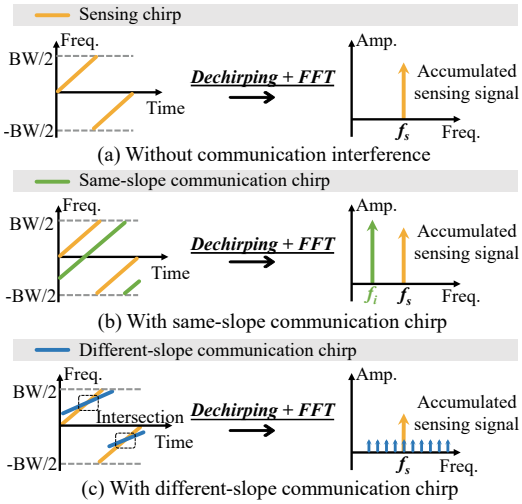


Figure 2: Two types of communication interference.

much more severely than it influences communication. Furthermore, we observe that the impact of communication interference on sensing is determined by the slopes of communication chirp and sensing chirp, and thus classify the communication interference into the following two types.

- *Same-slope communication chirp.* When the communication chirp and sensing chirp share the same slope, they exhibit similar time-frequency patterns as shown in Fig. 2(b). When accumulating the power of sensing chirp through dechirping and Fast Fourier transform (FFT), the power of the communication chirp is also concentrated. This causes multiple significant FFT peaks, leading to ambiguity in determining which peak corresponds to the accumulated sensing signal. This issue can be addressed by adopting existing communication techniques [7, 15, 25], which leverage the distinct time-domain and frequency-domain patterns exhibited by signals from different LoRa nodes to accurately identify the FFT peak of the accumulated sensing signal.
- *Different-slope communication chirp.* When the communication chirp and sensing chirp have different slopes, they intersect in both the frequency and time domains, as shown in Fig. 2(c). At these intersections, a portion of the communication chirp's power leaks into the sensing chirp. Since LoRa sensing relies on weak reflection signals compared to the stronger direct path signal, even this small power leakage can degrade sensing performance. In practice, it is highly likely for communication chirps to have different slopes. According to LoRa standard, there are 21 distinct configurations of spreading factor (SF) and bandwidth (BW) [33]. For example, for a sensing chirp with SF10 and BW125kHz, only two communication configurations share the same slope: (SF10, BW125kHz) and (SF12, BW250kHz). The remaining 19 configurations result in different-slope interference. This different-slope communication interference is typically ignored in LoRa communication, as its impact on communication is negligible [33, 34]. However, such different-slope interference can significantly affect sensing, calling for the development of a new solution.

Motivated by our findings, in this paper, we propose *InterSen* to enable contact-free LoRa sensing under communication interference. The core of *InterSen* is to address the power leakage caused by different-slope communication interference. We propose to design a novel power-scaling downchirp instead of employing the default downchirp for dechirping. Our basic idea is to minimize the contribution of the intersection parts in the accumulated sensing signal because the communication power leakage occurs at the intersection part. For the default downchirp, the amplitude is constant across all the signal samples. In our design, we assign zero amplitude to a portion of the downchirp, whose positions correspond to the intersection part of the sensing chirp. In this case, the accumulated sensing signal does not contain the power leakage from the communication chirp. While this power-scaling downchirp addresses the issue of power leakage from the communication chirp, it also introduces signal power variations due to the varying number of intersection parts in real-world scenarios. This power variation affects sensing, which relies on accurate signal amplitude (power) variations. To address this, we further scale the power of the designed downchirp to maintain constant power.

However, we observe that in occasional cases of severe communication power leakage, the power-scaling downchirp solution becomes less effective. This is due to the excessive number of intersection parts. After assigning zero amplitude to many parts of the downchirp, the power of the downchirp becomes very small and sensitive in the following power-scaling step, making it difficult to maintain a constant power for the designed downchirp. We observe that when the power of the intersection parts is more than 80% of the total chirp power, the performance of the proposed chirp design degrades. To overcome this problem, we propose to discard the sensing chirp when its intersection power exceeds 80%. To mitigate the impact of chirp discard, we adopt the technique of sparse reconstruction [9]. The intuition is that the sensing pattern, i.e., target-induced signal variations, is inherently sparse in the frequency domain. This means that the key information of the sensing pattern can be represented by a few frequency components, making it possible to reconstruct the pattern even with incomplete data. The main contributions are summarized as follows.

- We address a critical issue associated with contact-free LoRa sensing, i.e., the interference caused by LoRa communication on sensing. We believe this is a big step towards integrating LoRa sensing into the existing LoRa infrastructure.
- With a deep understanding of the impact of communication interference on sensing, we carefully design power-scaling downchirps, which not only remove the power leakage from the communication chirp but also ensure that the accumulated sensing signal remains effective for sensing. To handle the most severe cases, we discard the corrupted chirps and reconstruct the sensing pattern using the sparse reconstruction technique.
- We implement the prototype of *InterSen* on commodity LoRa nodes, and conduct comprehensive experiments with three representative sensing applications, i.e., respiration monitoring, walking distance estimation, and gesture recognition. Experiment results show that *InterSen* can enable LoRa sensing to work well in the presence of multiple interfering communication nodes.

2 PRELIMINARY

2.1 LoRa Communication

LoRa employs Chirp Spread Spectrum (CSS) modulation [33], which encodes the symbols using a series of upchirps. The base upchirp

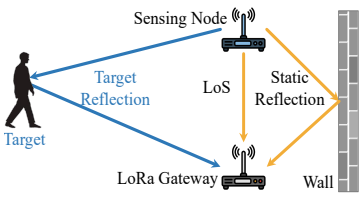


Figure 3: Modeling contact-free LoRa sensing.

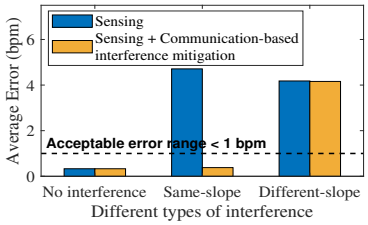


Figure 4: Sensing reliability under interference.

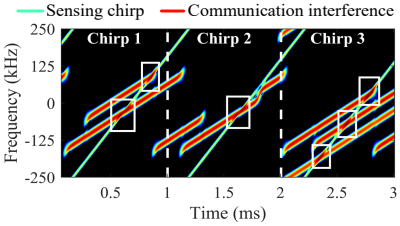


Figure 5: Sensing chirps with different intersections.

sweeps from $-\frac{BW}{2}$ to $\frac{BW}{2}$, where BW is the chirp bandwidth. Each symbol is represented by an upchirp with a unique initial frequency. The chirp frequency increases linearly over time at a rate $K = \frac{BW^2}{2^{SF}}$, where SF is the spreading factor. LoRa typically supports spreading factors (SF) ranging from 6 to 12 and bandwidths of 125 kHz, 250 kHz, and 500 kHz [33]. To demodulate the symbol, the LoRa gateway multiplies each upchirp with a base downchirp, i.e., dechirping. After FFT, all the signal samples are coherently added up at one frequency, and a peak can be obtained. The position of this peak, i.e., the frequency value, is used to decode the LoRa symbol.

2.2 Contact-free LoRa Sensing

Modeling LoRa sensing signal. Different from LoRa communication, contact-free LoRa sensing employs the target reflection signal to extract human motion information. The chirp transmitted by the sensing node is represented as

$$x_{\text{up}}(t) = e^{j2\pi(f_s t + \frac{K}{2}t^2 - \frac{BW}{2}t)}, \tag{1}$$

where f_s is the modulated frequency of the sensing chirp, and K is the chirp slope. The transmitted chirp signal propagates via multiple paths before arriving at the LoRa gateway. The received signal includes the line-of-sight (LoS) component and reflections from static objects (e.g., walls) and the human target. These components are grouped into: static signals, including the LoS and reflections from static objects (denoted as H_s) and dynamic signals, reflected from the moving human target, with time-varying channel (denoted as $H_d(t)$). Therefore, the received signal $R(t)$ at the gateway is

$$R(t) = \underbrace{e^{j2\pi(f_s t + \frac{K}{2}t^2 - \frac{BW}{2}t)}}_{\text{sensing chirp}} \cdot [H_d(t) + H_s]. \tag{2}$$

Accumulated sensing signal. The state-of-the-art LoRa sensing [27] exploits the chirp signal's unique properties to enable long-range sensing. Unlike ChirpSen [27], which accumulates sensing upchirps via time-domain convolution with a specific downchirp, we perform accumulation in the frequency domain using a fixed standard downchirp. Specifically, the received upchirp signal $R(t)$ is multiplied with a base downchirp $x_{\text{dn}}(t) = e^{j2\pi(-\frac{K}{2}t^2 + \frac{BW}{2}t)}$, which has a chirp slope opposite to that of the transmitted sensing chirp. This process, termed as dechirping, converts the time-varying chirp frequencies into a constant frequency f_s.

$$R_{dechirp}(t) = R(t) \cdot x_{\text{dn}}(t) = e^{j2\pi f_s t} \cdot [H_d(t) + H_s]. \tag{3}$$

As shown in Eq. (3), all the signal samples of $R_{dechirp}(t)$ share the same frequency f_s. After applying the FFT operation, the power of these signal samples can be accumulated at the frequency of f_s, and the accumulated sensing signal is

$$R_{accu}(f_s) = N \cdot [H_d(t) + H_s], \tag{4}$$

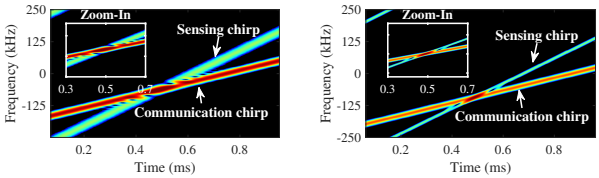
where N is the number of signal samples within the chirp, which is known in advance. During this process, the signal samples are constructively added up, while noise is randomly combined. This process significantly improves the power, and the accumulated sensing signal is then used to achieve a longer sensing range. By capturing the target-induced signal variation (i.e., sensing pattern) across a series of accumulated sensing signals, we can enable a variety of sensing tasks.

3 UNDERSTANDING COMMUNICATION INTERFERENCE ON SENSING

This section conducts a series of controlled experiments to investigate the impact of communication interference on sensing. Specifically, we deploy a single LoRa node to perform respiration sensing on a target located 13 m away from the gateway. Meanwhile, another LoRa node generates different types of communication interference. Without loss of generality, we employ the state-of-the-art LoRa sensing technique [27] to extract the amplitude variation for respiration sensing. Fig. 4 presents the respiration error rates, averaged over 50 experiment repetitions. Without interference, the target's respiration rate is captured accurately with an average error of 0.33 bpm (breaths per minute). On the other hand, communication interference results in errors as high as 4.71 bpm, far exceeding the acceptable performance threshold of 1 bpm. Moreover, we observe that the slope (i.e., the rate of frequency change) of the communication chirp affects the sensing performance.

Same-slope communication chirp. The same-slope communication chirp has similar time-frequency patterns as shown in Fig. 2(b). As a result, multiple FFT peaks appear in the frequency domain, leading to ambiguity—it is challenging to identify the correct peak corresponding to the accumulated sensing signal. Since existing LoRa sensing techniques overlook communication interference, they are prone to significant sensing errors. For example, the error significantly increases to 4.71 bpm in Fig. 4. For the first time, our preliminary experiments verify the feasibility of applying the technique of enabling concurrent communication on sensing under the same-slope communication interference. Specifically, we can exploit the unique patterns in the frequency or time domain to identify the FFT peak corresponding to the accumulated sensing signal [7, 15, 25]. Fig. 4 shows a respiration rate estimation error of 0.38 bpm after applying a representative concurrent communication technique (i.e., CIC [15]) to mitigate the effect of same-slope communication interference on sensing.

Different-slope communication chirp. When the communication chirp and the sensing chirp have different slopes, as shown in Fig. 2(c), the sensing chirp and the communication chirp have similar time-frequency patterns in the intersection area (marked by a rectangle). When we accumulate the sensing chirp, the signal samples of the communication chirp in the intersection part also get accumulated, resulting in the communication chirp leaking



(a) STFT spectrogram (b) CT spectrogram

Figure 6: Comparing two methods for time-frequency analysis.

power into the sensing chirp. This communication power leakage issue can change the accumulated sensing signal, including both amplitude and phase. One example is illustrated in Fig. 2: the accumulated sensing signal in Fig. 2(c) noticeably deviates from the one extracted without communication chirp in Fig. 2(a).

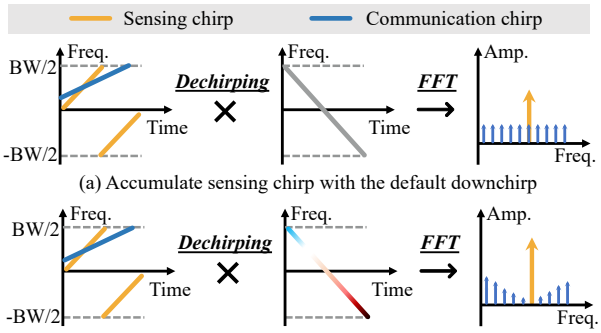
More importantly, this distortion easily destroys the reflection signal used for sensing. This is because the reflection signal is reflected by the sensing target and is inherently very weak. The dominating component of the accumulated sensing signal is the LoS signal, which does not contain the information of the sensing target and can not be utilized for sensing. Fig. 4 shows that different-slope interference leads to a respiration error of 4.18 bpm, even though the power leakage caused by interference is low (e.g., 7 dB lower than the accumulated sensing signal in our preliminary experiment).

Note that this different-slope interference is not a problem in LoRa communication since LoRa communication relies on signal frequency for data decoding and the frequency of the accumulated signal (i.e., peak location) does not change in the presence of other chirps as shown in Fig. 2(c). Fig. 4 shows that existing concurrent communication techniques cannot mitigate the respiration sensing error under different-slope interference.

Challenges in addressing different-slope interference. The fundamental difference between sensing and communication requires a new solution to address the impact of different-slope communication chirp on sensing chirp. This is challenging because the number and locations of intersections vary, as shown in Fig. 5, causing each sensing chirp to be affected differently by communication interference. Moreover, in practice, there is a high chance that the communication chirp has a different slope. According to the LoRa standard, there are 21 different (SF, BW) configurations [33]. For a single sensing chirp configuration (e.g., SF10, BW125 kHz), the probability of different slope interference is 90.5%, since out of 21 communication chirp configurations, 19 have different chirp slopes.

4 SYSTEM DESIGN

This section introduces *InterSen*, which aims to enable accurate LoRa sensing under different-slope communication interference. To achieve this goal, *InterSen* consists of three key components: intersection detection, downchirp power scaling, and sensing pattern reconstruction. For each sensing chirp, *InterSen* identifies its intersection region with communication chirps, where power leakage occurs, and applies power scaling to adjust the downchirp used in sensing chirp accumulation, thus effectively eliminating the power leakage of communication interference. Under extremely severe communication power leakage, the proposed power-scaling downchirp solution becomes less effective. In such cases, *InterSen* discards the corrupted sensing chirps and adopts sparse reconstruction, which exploits the inherent sparsity of the sensing pattern in the frequency domain to recover the sensing information.



(a) Accumulate sensing chirp with the default downchirp

(b) Accumulate sensing chirp with the power-scaling downchirp

Figure 7: Basic idea of downchirp power scaling.

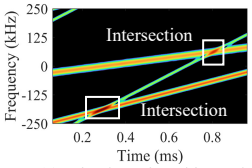
4.1 Intersection Detection

This section identifies the intersection parts between sensing and communication chirps by analyzing them in the time-frequency domain. A common time-frequency analysis approach is the Short-Time Fourier Transform (STFT) [11, 14, 17], which relies on the fixed-frequency sinusoidal basis functions and calculates their correlation with mixed chirps in successive time windows [38]. However, STFT has a limited resolution to accurately identify intersection segments. This is because STFT employs fixed-frequency sinusoidal functions and is difficult to adapt to the rapid frequency change of LoRa chirps, leading to inaccuracy in representing the intersection parts. As shown in Fig. 6(a), the intersection part in the STFT spectrogram is blurred, making it difficult to accurately identify the start and end points of the intersection.

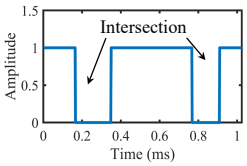
To accurately identify the intersection parts, we adopt a chirplet-based technique [35]. Different from STFT, the chirplet transform employs a set of chirp templates as basis functions. The chirplet transform process can be represented as

$$\mathrm{CT}(t,f) = \int x(\tau)\psi(\tau - t, f, K_s)d\tau, \tag{5}$$

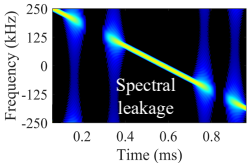
where $\psi(t, f, K_s) = e^{j(2\pi f t + \pi K_s t^2)}$ is the base function employed in our design, and K_s is the chirp slope, which is the same as that of sensing chirp. This base function aligns with the frequency variation trend of the LoRa chirp, enabling more precise tracking of the chirp feature in the time and frequency domains. As shown in Fig. 6(b), this method effectively eliminates the blur part observed in the STFT spectrogram and obtains a clearer representation of the intersection part. Based on the chirplet transform spectrogram, we identify the intersection part with one key observation: the amplitude of the sensing chirp in the intersection part exhibits significant variation even when communication interference is weak (e.g., 9 dB weaker than the sensing signal), while its amplitude remains relatively stable in the rest parts. This occurs because the communication chirp contaminates the intersection part, while the rest of the sensing chirp remains clean. Specifically, we track the normalized amplitude of the sensing chirp in the spectrogram and also apply a Gaussian smoothing filter [3] to mitigate noise in low SNR scenarios. After smoothing, we use a variance-based sliding window to identify the start and end points of the intersection, where the variance exceeds a pre-defined threshold. In *InterSen*, the window size is set to $2^{SF}/16$, and the variance threshold is 0.002 empirically. Our experiments show a 96% success rate for intersection detection across diverse settings.



(a) Mixed sensing chirp and communication chirps

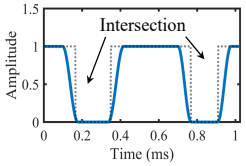


(b) Assign the zero amplitude to a part of the downchirp

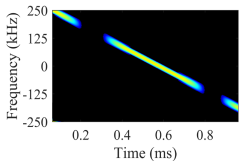


(c) Spectrogram of simple power-scaling downchirp

Figure 8: The straightforward design for the power-scaling downchirp.

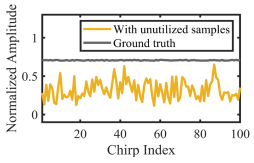


(a) Change the amplitude of power-scaling downchirp smoothly

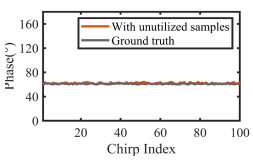


(b) Spectrogram of updated power-scaling downchirp

Figure 9: Changing the amplitude of downchirp smoothly.

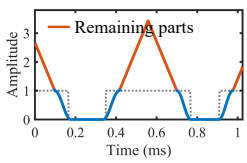


(a) Amplitude of the accumulated sensing signal

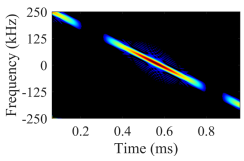


(b) Phase of the accumulated sensing signal

Figure 10: Impact of the varying number of unutilized signal samples on the accumulated sensing signal.



(a) Increase the amplitude of downchirp's remaining parts



(b) Spectrogram of final power-scaling downchirp

Figure 11: Increasing the amplitude of the power-scaling downchirp's remaining parts.

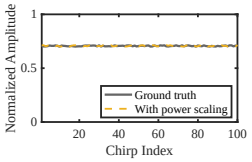


Figure 12: Amplitude of the accumulated sensing signal after careful power scaling.

4.2 Downchirp Power Scaling

To remove the impact of communication interference on the sensing chirp, we design a power-scaling downchirp to replace the default downchirp during the sensing signal accumulation.

4.2.1 ***Basic Idea.*** The proposed power scaling scheme aims to minimize the impact of intersections in the accumulated sensing signal because communication power leakage occurs there. To achieve this, instead of using the default downchirp with constant amplitude in Fig. 7(a) for dechirping, we design a power-scaling downchirp with varying signal power, as shown in Fig. 7(b), where different colors represent distinct power. In the default downchirp, all signal samples contribute equally during sensing signal accumulation. In contrast, power-scaling downchirp assigns zero amplitude to part of the downchirp (highlighted in white), whose positions correspond to the intersections at the sensing chirp, while assigning higher amplitude (highlighted in blue and red) to the remaining part. In this case, the sensing chirp remains accumulated at the same frequency point, and the accumulated sensing signal does not contain the power leakage from communication chirp. Although the idea is straightforward, designing an effective power-scaling downchirp is non-trivial and requires addressing two key challenges.

- *Additional spectral leakage interference.* Abrupt amplitude change can cause additional spectral leakage. As shown in Fig. 8(b) and (c), a straightforward design for the new downchirp is to assign zero amplitude to the intersection-corresponding part of the downchirp while maintaining the original amplitude (i.e., 1) for the remaining part. However, the abrupt amplitude shifts from 1 to 0 and back to 1 introduce sharp discontinuities in the new downchirp, resulting in spectral leakages as highlighted in blue in Fig. 8(c). When applying this new downchirp during the dechirping process, we find that spectral leakage induces additional interference to the accumulated sensing signal.
- *Random variations in the accumulated sensing signal.* Recall that the purpose of applying downchirp is to align all the signal samples of the sensing chirp and to make them constructively combined. This process significantly improves the signal power, and the power of the accumulated sensing signal is determined by the number of signal samples in the sensing chirp. However, the zero amplitude of the new downchirp prevents the intersection part of the sensing chirp from contributing to the accumulated sensing signal. As each sensing chirp intersects differently with communication chirps, as shown in Fig. 5, the number of unutilized signal samples (i.e., unutilized intersection) in each sensing chirp fluctuates. This fluctuation leads to random amplitude variations in different accumulated sensing signals, as illustrated in Fig. 10(a).[1] Note that the phase value of the accumulated sensing signal remains unchanged because it is determined by the phase values of the sensing chirp and downchirp, both of which do not change, as shown in Fig. 10(b). These combined effects render the distorted accumulated sensing signal unusable for sensing.

In the following sections, we carefully design power-scaling downchirps, which not only remove the communication power leakage but also ensure that the accumulated sensing signal remains effective for sensing.

4.2.2 ***Addressing Additional Spectral Leakage Interference.*** To avoid additional spectral leakage, we propose to change the amplitude of the downchirp smoothly between the default value and zero value. Specifically, compared to Fig. 8(b), Fig. 9(a) shows a more smoothly graduated curve between the default amplitude and zero amplitude. The corresponding downchirp is shown in Fig. 9(b), which does not exhibit the issue of spectral leakage. To achieve this objective, we scale the amplitude based on a cosine function

$$A_{\text{down}}(t) = \frac{1}{2} - \frac{1}{2}\cos\left(\pi \cdot \frac{t}{\Delta T_1}\right), \quad t \in \Delta T_1 \tag{6}$$

where ΔT_1 represents the duration of the smooth amplitude transition, which is set as half the corresponding intersection part. Note that for the part of downchirp, whose positions correspond to the intersection part at the sensing chirp, its amplitude is set as zero,

$$A_{\text{down}}(t) = 0, \quad t \in \Delta T_2, \tag{7}$$

where ΔT_2 represents the duration of intersection part.

4.2.3 ***Addressing Random Variations in Accumulated Sensing Signal.*** To compensate for the random amplitude variations, we propose to assign a larger amplitude to the remaining downchirp. Specifically, we apply the following equation to assign the amplitude of the remaining downchirp

$$A_{\text{down}}(t) = 1 + k \cdot d(t), \quad t \in \Delta T_3, \tag{8}$$

[1]The ground truth is obtained when there is no communication interference.

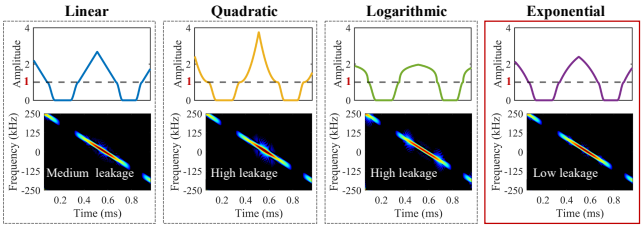
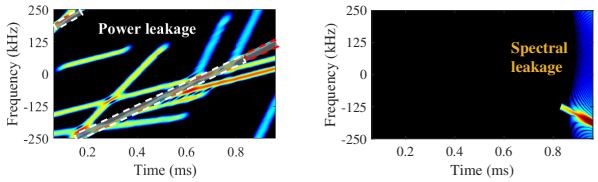


Figure 13: Power-scaling downchirp with different scaling functions.



(a) Severe power leakage (b) Ineffective power-scaling downchirp

Figure 14: An example of power scaling under severe power leakage.

where ΔT_3 denotes the duration of the remaining downchirp, $d(t)$ indicates how close the current time t is to the boundary of the transition periods (i.e., ΔT_1), and k is a constant determined by the amplitude loss resulting from the spectral leakage mitigation process. As shown in Fig. 11(a), the amplitude of the remaining downchirp is higher than the default value. Fig. 11(b) presents the updated downchirp.

Interestingly, besides linearly scaling the amplitude for the remaining downchirp, we observe there are non-linear ways to scale the amplitude, including quadratic, logarithmic, and exponential. Table 1 summarizes these functions. Without loss of generality, each function includes two parameters: (1) $d(t)$ represents how close the current time t is to the boundary of the transition periods (i.e., ΔT_1), and (2) k determines how the amplitude is scaled, e.g., linear or non-linear. Fig. 13 presents power-scaling downchirps with these scaling functions. We observe that downchirps with quadratic, logarithmic, and linear scaling functions have the spectral leakage issue, while the downchirp with exponential scaling function has little spectral leakage. To evaluate the impact of these induced spectral leakages during the amplitude compensation, we collect 10,000 sensing chirps with various amounts of communication power leakage. As shown in Table 1, the exponential scaling function demonstrates the best performance, inducing minimum error to the amplitude of the accumulated sensing signal. Therefore, we employ the exponential function to scale the amplitude of the remaining downchirp. With such a design, the random power deduction in Fig. 10(a) is addressed, and the accumulated sensing signal demonstrates a stable amplitude variation in a static environment, as shown in Fig. 12.

In summary, by combining Eq. 6, Eq. 7, and Eq. 8, we can design an effective downchirp whose amplitude varies over the chirp duration for a specific sensing chirp to eliminate the impact of communication interference on it.

Upper limit of power-scaling downchirp. Our benchmark experiments show the sensing reliability via our power-scaling downchirp scheme remains high across a wide range of settings. However, in cases with a large amount of communication power leakage, the power-scaling downchirp solution becomes less effective. One example is shown in Fig. 14(a), where around 80% of the sensing chirp is intersected with communication chirps. In this case, as shown in Fig. 14(b), after assigning zero amplitude to a large portion of the downchirp, the power of the downchirp becomes very small, and the subsequent power-scaling step induces severe spectral leakage. We also perform benchmark experiments by varying the level of communication power leakage to determine the threshold at which the power-scaling downchirp design becomes less effective. Fig. 15 depicts the statistical results. If the intersection ratio (i.e., the ratio of intersection parts to the whole sensing chirp) is less than 80%, our technique can accurately obtain the accumulated sensing signal, with amplitude error as small as 0.01. However, when intersection ratio exceeds 80%, the error increases. We explain how to address this issue in Section 4.3.

Table 1: Comparison of different scaling functions.

	Function	Leakage degree	Mean amplitude error
Linear	$1 + kd(t)$	Medium	0.02
Quadratic	$1 + kd(t)^2$	High	0.05
Logarithmic	$1 + k\log_{10}(1 + d(t))$	High	0.05
Exponential	$1 + k\left(1 - e^{\frac{-d(t)}{\max(d(t))}}\right)$	**Low**	**0.01**

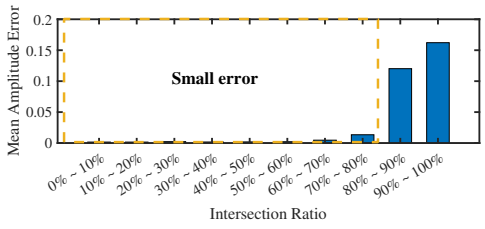


Figure 15: Power-scaling downchirp design varies with intersection ratio.

4.3 Sensing Pattern Reconstruction

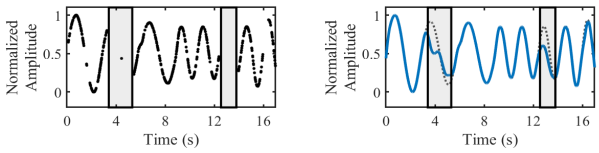
Since communication interference is dynamic, each sensing chirp is affected by varying levels of interference as shown in Fig. 5. We discard those severely contaminated sensing chirps and leverage the property of sparsity in the frequency domain to reconstruct the complete sensing pattern for reliable sensing.

4.3.1 ***Sparse Reconstruction.*** Our idea is to adopt the sparse reconstruction [9], which is a powerful tool for signal recovery to recover the complete sensing pattern. Our key intuition is that the sensing pattern, i.e., target-induced signal variation, is inherently sparse in the frequency domain. It means that the key information of the sensing pattern can be represented by a few frequency components, making it possible to reconstruct the pattern even with incomplete data. The inherent sparsity arises from the fact that human activities, such as respiration, walking, and hand gestures, are confined to a narrow frequency range, typically below 20 Hz [8]. For example, human respiration rates generally fall within 0.2 Hz to 0.5 Hz [36].

To reconstruct the sensing pattern, we formulate the problem as a sparse signal recovery task in the frequency domain. Specifically, we solve the following optimization problem

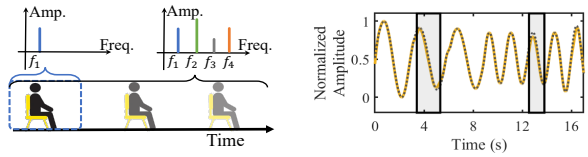
$$\min ||S_{tar}||_1 \quad \text{s.t.} \quad M \cdot s_{raw} = S_{tar}, \tag{9}$$

where s_{raw} is the observed raw sensing pattern, which is incomplete and M is the Fourier transform matrix. Note that $M \cdot s_{raw}$ represents the observed raw sensing pattern in the frequency domain, which is known in advance. S_{tar} represents the desired complete sensing pattern that we aim to obtain, expressed in the frequency domain. We adopt the interior point method-based algorithm [9] to solve



(a) Incomplete sensing pattern (b) Sparse reconstruction result

Figure 16: An example of sparse reconstruction.



(a) Frequency components vary with time (b) Multi-scale refinement result

Figure 17: Illustration of multi-scale refinement.

the above ℓ_1-norm minimization problem, which derives a sensing pattern that best matches the observed one. Finally, we perform the Inverse Fourier Transform (IFFT) on the derived s_{tar} to obtain the reconstructed sensing pattern in the time domain.

4.3.2 ***Multi-scale Refinement.*** After applying the sparse reconstruction scheme, we observe that parts of the reconstructed sensing pattern might be inaccurate when the observed samples are insufficient (e.g., the area marked by the gray rectangle in Fig. 16(a)). As shown in Fig. 16(b), this can lead to significant reconstruction errors. This issue arises because sparse reconstruction relies on the frequency-domain sparsity of the sensing pattern over the entire observation period. When observed samples in a local region are insufficient, the optimization process in Eq. 9 may introduce some irrelevant frequency components from other regions to fill in gaps.

To address these local inaccuracies caused by insufficient samples, we propose multi-scale refinement. Our key observation is that the frequency components of the sensing pattern within a short duration are generally more concentrated than those over the entire time frame, as shown in Fig. 17(a). In this case, sparse reconstruction performed over local durations may concentrate on the relevant frequency components specific to those regions, avoiding introducing unrelated frequencies.

The refinement process includes three steps. First, we apply the method in Section 4.3.1 to obtain an initial sensing pattern. Next, we identify local regions with insufficient samples using a sliding window (i.e., 0.5 s). We compute the proportion of observed samples in each window and mark those below a predefined threshold (5% empirically). Consecutive marked windows are merged to determine regions requiring refinement. For each identified region, we define windows of varying sizes and use observed samples in each window for local sparse reconstruction. The windows are centered on the region, and the sizes are determined by the region's length multiplied by a set of scaling factors. (1.5, 2, 2.5, 3, 3.5 in our setup)

$$\min ||S_w||_1 \quad \text{s.t.} \quad M_w \cdot s_w = S_w, \tag{10}$$

where S_w and $M_w \cdot s_w$ are the reconstructed and observed sensing pattern in the frequency domain within a window, respectively. This multi-scale approach allows us to adaptively capture local frequency features at different time scales. A small window can more accurately reflect local frequency components, while a large window can utilize more observed samples for reconstruction. Finally, we calculate the reconstruction error by comparing the observed samples within each window to the reconstructed values. The window that yields the lowest error is selected to refine the initial sensing pattern. As shown in Fig. 17(b), this multi-scale local refinement enhances sensing pattern reconstruction accuracy during periods with limited observed samples.

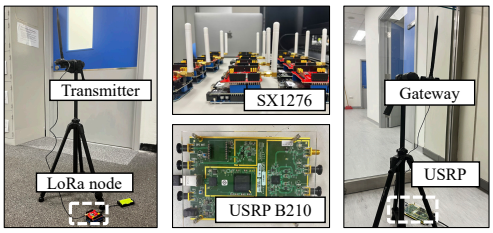


Figure 18: Hardware devices.

5 EVALUATION

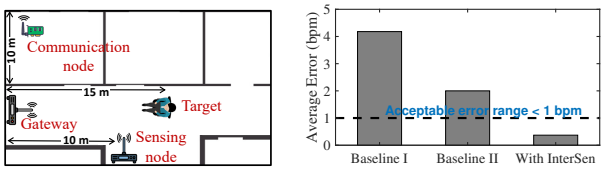
5.1 Implementation

Hardware. We implement *InterSen* using a USRP B210 as a gateway and commodity LoRa nodes as transmitters. Fig. 18 shows the hardware devices. The LoRa node is composed of a Dragino LoRa shield embedded with a Semtech SX1276 module. We connect the LoRa shield to an Arduino Uno board to configure the chirp parameters. The LoRa node operates at the 915 MHz band with a transmission power of 20 dBm. The sampling rate of the gateway is set to 500 kHz. Both the LoRa node and gateway are equipped with a VERT900 omni-directional antenna with a gain of 3 dBi. *InterSen* does not require any hardware modifications to the LoRa nodes.
Software. We use GNU Radio software to collect LoRa signals and implement *InterSen* in MATLAB to process the signals. Since contact-free sensing does not care about the transmitted data but the target-induced signal variation, the sensing signal is a series of identical and known chirps in our evaluation. When the LoRa sensing signal is received by the gateway, we first remove the time and frequency offsets using the same methods in ChirpSen [27] to ensure synchronization between the LoRa node and the gateway. All the signal processing happens on a Lenovo Thinkstation P360 tower desktop with an Intel Core i7-12700 CPU and 64 GB RAM.

5.2 Benchmark Experiments

This section evaluates *InterSen*'s two key design components, i.e., downchirp power scaling (Section 4.2) and sensing pattern reconstruction (Section 4.3).

Experiment settings. We conduct experiments in a workplace as shown in Fig. 19(a). One LoRa node (i.e., sensing node) performs respiration monitoring, and the other node (i.e., communication node) generating communication interference. We evaluate the performance with more nodes in the next two sections. The distance between the target and gateway is set to 15 m, and the distance between the sensing node and gateway is 10 m. The communication node is deployed 10 m away from the gateway. The spreading factor (SF) and the bandwidth (BW) of the sensing node are set to 10 and 500 kHz, while the communication node operates with SF 8 and BW 500 kHz. We capture the amplitude variations induced by respiration for around 20 s as the sensing pattern and calculate the respiration rate using FFT. The ground truth is obtained from a commodity device, i.e., HEXOSKIN Smart Garments. The experiment is repeated 30 times to calculate the average respiration error.



(a) Experiment scenario (b) Overall performance

Figure 19: Comparison experiments with two baselines.

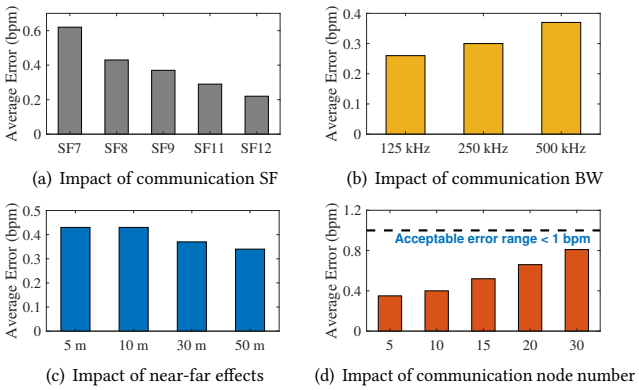
Performance evaluation. We compare *InterSen* with two baselines. (1) Baseline I: traditional LoRa sensing techniques [27] without considering communication interference. (2) Baseline II: power scaling downchirp design + interpolation. Specifically, this baseline adopts power scaling to suppress communication power leakage. Since power scaling becomes inaccurate when power leakage is too severe, it adopts spline interpolation [13] to interpolate missing parts. Fig. 19(b) shows the performance comparison. Baseline I, which does not address communication interference, yields an average respiration estimation error of 4.2 bpm. While Baseline II incorporates power scaling to mitigate communication power leakage, some accumulated sensing signals are still invalid because of severe power leakage from communication interference. Although this baseline adopts interpolation, it suffers from inaccurate reconstruction of the sensing pattern, resulting in an error of 1.9 bpm. In contrast, *InterSen* adopts sparse reconstruction and multi-scale refinement to recover the complete sensing pattern, achieving a low error of 0.38 bpm in the presence of communication interference.

5.3 Impact of Different Parameters

This section evaluates the impact of different parameter settings on *InterSen*. The experiment is still conducted in the workplace as shown in Fig. 19(a) and respiration sensing is employed as the application example. The locations of the gateway, sensing node, and target, as well as the parameters of the sensing node remain unchanged. We explore the system performance by changing the parameter configuration, location, and number of communication nodes. Each experiment is repeated 30 times.

5.3.1 ***Impact of Communication SFs.*** Spreading factor (SF) is a crucial LoRa chirp parameter affecting chirp duration and noise resistance. In this experiment, we change the SF of the communication node to 7, 8, 9, 11, and 12, to evaluate its impact on *InterSen*. SF=10 is excluded as it causes interference with the same slope as the sensing chirp, which can be mitigated by existing algorithms (e.g., [15, 34]). As shown in Fig. 20(a), *InterSen* effectively suppresses interference, achieving accurate respiration monitoring (i.e., error < 0.7 bpm). However, the performance decreases as the SF of communication node decreases. This is because a smaller SF results in shorter communication chirps, resulting in more communication chirps and more intersection parts with sensing chirp. For example, with the sensing node using SF=10 and the communication node using SF=7, up to 8 communication chirps can intersect with a single sensing chirp, leading to more communication power leakage. Despite this, the average error of respiration rate estimation of *InterSen* remains within an acceptable range (0.62 bpm for SF=7).

5.3.2 ***Impact of Communication Bandwidths.*** Bandwidth (BW) is another key LoRa parameter. This experiment varies the communication node's BW from 125 kHz, 250 kHz, to 500 kHz, with the SF fixed as 9. Fig. 20(b) shows the average error of respiration rate estimation under different communication BW settings. *InterSen* effectively addresses the interference issue across different BWs, ensuring accurate respiration monitoring. However, as BW increases, the performance slightly declines, with errors of 0.26 bpm, 0.30 bpm, and 0.37 bpm at 125 kHz, 250 kHz, and 500 kHz, respectively. This is because larger BW reduces chirp duration, leading to more communication chirps intersecting with the sensing chirp.



(a) Impact of communication SF (b) Impact of communication BW

(c) Impact of near-far effects (d) Impact of communication node number

Figure 20: Impact of different parameter settings on *InterSen*.

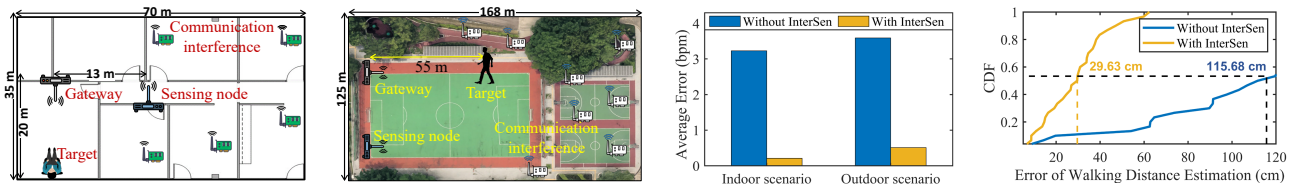
5.3.3 ***Impact of Near-Far Effects.*** In this experiment, we vary the distance between the communication node and the gateway (5 m, 10 m, 30 m, and 50 m), while keeping the distance between the sensing node and gateway unchanged, i.e., 10 m, to study the impact of near-far effects on the performance. The sensing node's parameters remain SF=10 and BW=500 kHz, while the communication node is set to SF=8 and BW=500 kHz. In this case, the power ratio between the sensing signal and the communication signal varies. As shown in Fig. 20(c), *InterSen* maintains robustness under different distances. Even when the communication interference source is just 5 m away from the gateway, it still achieves a low respiration error of 0.44 bpm.

5.3.4 ***Impact of the Number of Communication Nodes.*** We increase the number of communication nodes to verify the scalability of *InterSen*. Specifically, we place more communication nodes in three classrooms as shown in Fig. 19(a). Each communication node selects different SF and BW configurations based on its distance from the receiver, with SFs set to 7, 8, 9, 11, and 12, and BWs to 125 kHz, 250 kHz, and 500 kHz. An additional beacon node coordinates transmissions to generate collisions. Upon receiving a beacon, the sensing node sends sensing signal continuously at 100% duty cycle. Communication packets arrive at the gateway with different time offsets. The transmission interval for each communication node is randomized between 0 and 3 s. Fig. 20(d) shows that *InterSen* maintains acceptable error levels as the number of communication nodes increases. Even in the most challenging scenario with 30 communication nodes, it remains reliable, with a small error of 0.81 bpm.

5.4 Case Study

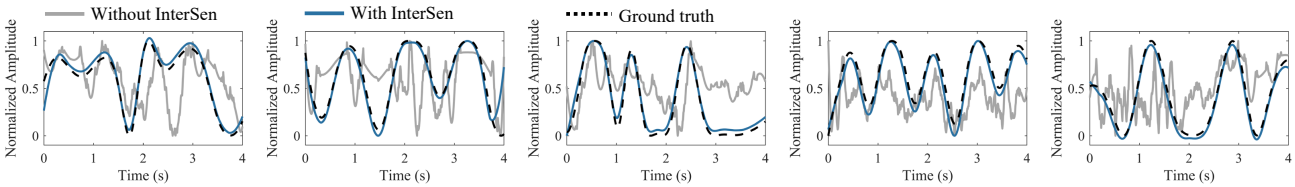
This section evaluates *InterSen* with three real-world applications in both indoor (35 m × 70 m) and outdoor scenarios (125 m × 168 m).

Experiment scenarios. We evaluate *InterSen* in two real contact-free LoRa sensing scenarios. Fig. 21(a) shows the indoor scene with 6 LoRa nodes. The distance between the target and the gateway is



(a) Indoor scenario (b) Outdoor scenario (c) Respiration sensing performance (d) Walking sensing performance

Figure 21: Respiration sensing and walking distance estimation under communication interference.



(a) Wave hand (b) Push and pull (c) Draw a circle (d) Clap hand (e) Swing arm

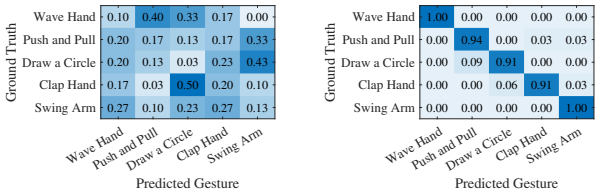
Figure 22: Examples of gesture sensing patterns without and with *InterSen*.

20 m. The sensing node is 13 m from the gateway. The remaining five LoRa nodes generate communication interference. Fig. 21(b) shows the outdoor scenario with 9 LoRa nodes. The sensing node is 30 m from the gateway. The initial distance between the target and gateway is 55 m.

5.4.1 ***Case Study I: Respiration Sensing.*** We conduct experiments in both indoor and outdoor scenarios, repeating each experiment 30 times to calculate the average error in respiration rate estimation. As shown in Fig. 21(c), without *InterSen*, the respiration sensing performance is poor under communication interference. The average error of respiration rate estimation is 3.23 bpm and 3.59 bpm in the indoor and outdoor scenarios, respectively. *InterSen* reduces the errors to 0.21 bpm indoors and 0.50 bpm outdoors, representing improvements of 93.5% and 86.1%, respectively.

5.4.2 ***Case Study II: Walking Distance Estimation.*** Estimating human walking distance without attaching any sensors to the individual is an interesting topic in contact-free tracking. We conduct this experiment in the outdoor scenario as shown in Fig. 21(b). In each experiment, we capture the amplitude variations of accumulated sensing signal induced by walking, and estimate the walking distance by counting the number of variation cycles [27]. We compare the estimated walking distance with the ground truth obtained using a Xunta AK-800 laser rangefinder. We conduct 30 experiments with varying walking distances ranging from 1 to 6 m. The overall cumulative distribution function (CDF) of walking distance estimation errors is shown in Fig. 21(d). The median error without applying *InterSen* is 115.7 cm, while with *InterSen*, the median error is reduced to 29.6 cm, representing a 74.4% improvement.

5.4.3 ***Case Study III: Gesture Recognition.*** Gesture recognition is an important application of wireless sensing, where different gestures are identified by analyzing the sensing patterns. We evaluate gesture recognition performance in the indoor scenario in Fig. 21(a). In this experiment, we consider five distinct gestures: wave hand, push and pull, draw a circle, clap hand, and swing arm. The distance between the target and gateway is 7 m. Firstly, we collect the clean sensing patterns of all gestures without any interference to establish ground truths. Then, we collect sensing patterns under communication interference and apply *InterSen*. Each gesture is performed 30 times, and a classic recognition algorithm [10] is used to classify the gestures. Fig. 22 shows an example of different



(a) Without *InterSen* (b) With *InterSen*

Figure 23: Confusion matrix of predicted gesture results.

gesture sensing patterns. Without applying *InterSen*, the sensing patterns deviate substantially from the ground truth. In contrast, the patterns processed with *InterSen* closely align with the ground truth. Fig. 23 presents the confusion matrix for gesture recognition without and with *InterSen*. *InterSen* achieves over 91% accuracy across all tested gestures in this experiment.

6 DISCUSSION

Sensing range. *InterSen* achieves a sensing range of 55 m in Section 5.4.2, which is significantly larger than other sensing modalities (e.g., 3-8 m for WiFi sensing), demonstrating the potential long-range sensing capability. However, since contactless sensing relies on weak reflection signals, LoRa sensing distance is inherently smaller than LoRa communication distance (kilometers), which has been discussed in prior LoRa sensing literature [27, 31, 36].

Multi-target sensing. Multi-target sensing is a well-known challenge in contact-free sensing. When multiple targets are present in the same environment, their reflection signals mix at the receiver, making it difficult to extract individual movement information [28, 31, 36]. We conduct experiments in a single-target scenario because *InterSen* focuses on addressing the impact of communication interference on sensing. We will explore multi-target sensing in the presence of communication interference in our future work.

7 RELATED WORK

Contact-free LoRa sensing. LoRa sensing has been studied over the past few years [2, 27, 29–31, 36, 37]. These works aim to improve the sensing range and through-wall capability of LoRa for apartment-level and even building-level human sensing in a contact-free manner. Although promising, these systems assume there is no LoRa communication interference.

Concurrent LoRa communication. Concurrent LoRa communication has been extensively explored [4, 7, 12, 15, 19, 20, 22, 25, 26, 32, 33]. These studies typically focus on communication-to-communication interference and analyze the unique signal characteristics to address same-slope communication interference, ensuring accurate demodulation. However, communication-to-sensing interference, specifically different-slope communication interference, also significantly distorts sensing, and *InterSen* introduces innovative solutions to tackle this issue.

Integrated sensing and communication. Integrated Sensing and Communication (ISAC) has recently gained significant interest from both academia and industry. This concept has been extensively studied in customized radar systems [5]. Recently, SenCom [6] utilizes general WiFi traffic for sensing, however, it relies on beamforming via multiple antennas and is infeasible for low-end LoRa end devices. LoSense [16] tracks the location of LoRa nodes while maintaining communication capabilities, while it is not designed for contact-free LoRa sensing. LSencom [28] enables LoRa sensing to coexist with communication. However, it requires the sensing node to transmit a special sensing chirp that deviates from the standard LoRa waveform, thereby preventing the reuse of LoRa communication packets for sensing purposes. In contrast, *InterSen* maintains compatibility with the existing LoRa waveform and can leverage existing LoRa communication packets for sensing. When receiving a standard LoRa packet, it demodulates transmitted symbols, and accumulates the sensing signal from each standard LoRa chirp, and mitigates communication interference. By doing so, *InterSen* can be seamlessly integrated into the LoRa communication infrastructure, laying a solid foundation for next-generation ISAC.

8 CONCLUSION

In this paper, we propose *InterSen*, a novel solution for addressing communication interference in LoRa sensing. By modeling the impact of communication interference on sensing, we develop a power-scaling downchirp design and a sensing pattern reconstruction scheme. Extensive evaluations show the effectiveness and generalization of *InterSen*. We believe this is a big step towards the adoption of the LoRa network for both sensing and communication.

9 ACKNOWLEDGMENTS

This work was partially supported by CityU 11206023, ECS CityU 21216822, CRF C1045-23G, NSFC Grant (No. 62372307, No. U2001207), Guangdong NSF (No. 2024A1515011691), and NTU SUG-NAP.

REFERENCES

[1] Zhaoxin Chang, Fusang Zhang, Jie Xiong, Weiyan Chen, and Daqing Zhang. 2024. MSense: Boosting Wireless Sensing Capability Under Motion Interference. In *ACM Mobicom*.
[2] Lili Chen, Jie Xiong, Xiaojiang Chen, Sunghoon Ivan Lee, Kai Chen, Dianhe Han, Dingyi Fang, Zhanyong Tang, and Zheng Wang. 2019. WideSee: towards wide-area contactless wireless sensing. In *ACM SenSys*.
[3] Marc Peter Deisenroth, Ryan Darby Turner, Marco F. Huber, Uwe D. Hanebeck, and Carl Edward Rasmussen. 2012. Robust Filtering and Smoothing with Gaussian Processes. *IEEE Trans. Automat. Control* (2012).
[4] Rashad M. Eletreby, Diana Zhang, Swarun Kumar, and Osman Yağan. 2017. Empowering Low-Power Wide Area Networks in Urban Settings. In *ACM SIGCOMM*.
[5] Liang Han and Ke Wu. 2013. Joint wireless communication and radar sensing systems–state of the art and future prospects. *IET Microwaves, Antennas & Propagation* (2013).
[6] Yinghui He, Jianwei Liu, Mo Li, Guanding Yu, Jinsong Han, and Kui Ren. 2023. SenCom: Integrated Sensing and Communication with Practical WiFi. In *ACM MobiCom*.
[7] Bin Hu, Zhimeng Yin, Shuai Wang, Zhuqing Xu, and Tian He. 2020. SCLoRa: Leveraging Multi-Dimensionality in Decoding Collided LoRa Transmissions. In *IEEE ICNP*.
[8] Bryce Kellogg, Vamsi Talla, and Shyamnath Gollakota. 2014. Bringing gesture recognition to all devices. In *USENIX NSDI*.
[9] Jihun Kim, Qingfei Wang, Siyuan Zhang, and Sangpil Yoon. 2021. Compressed Sensing-Based Super-Resolution Ultrasound Imaging for Faster Acquisition and High Quality Images. *IEEE Transactions on Biomedical Engineering* (2021).
[10] Anna Lekova and Mo Adda. 2015. Hand Gesture Recognition Based on Signals Cross-Correlation. *Science Gate Publishing* (2015).
[11] Chenning Li, Hanqing Guo, Shuai Tong, Xiao Zeng, Zhichao Cao, Mi Zhang, Qiben Yan, Li Xiao, Jiliang Wang, and Yunhao Liu. 2021. NELoRa: Towards Ultra-low SNR LoRa Communication with Neural-enhanced Demodulation. In *ACM SenSys*.
[12] Chenning Li, Xiuzhen Guo, Longfei Shangguan, Zhichao Cao, and Kyle Jamieson. 2022. CurvingLoRa to Boost LoRa Network Throughput via Concurrent Transmission. In *USENIX NSDI*.
[13] Sky McKinley and Megan Levine. 1998. Cubic spline interpolation. *College of the Redwoods* (1998).
[14] Kanav Sabharwal, Soundarya Ramesh, Jingxian Wang, Dinil Mon Divakaran, and Mun Choon Chan. 2024. Enhancing LoRa Reception with Generative Models: Channel-Aware Denoising of LoRaPHY Signals. In *ACM SenSys*.
[15] Muhammad Osama Shahid, Millan Philipose, Krishna Chintalapudi, Suman Banerjee, and Bhuvana Krishnaswamy. 2021. Concurrent Interference Cancellation: Decoding Multi-Packet Collisions in LoRa. In *ACM SIGCOMM*.
[16] Z. Song, S. Tong, and J. Wang. 2023. LoSense: Integrated Long-Range Sensing and Communication with LoRa Signals. In *IEEE ICNP*.
[17] Kai Sun, Zhimeng Yin, Weiwei Chen, Shuai Wang, Zeyu Zhang, and Tian He. 2021. Partial Symbol Recovery for Interference Resilience in Low-Power Wide Area Networks. In *IEEE ICNP*.
[18] Zehua Sun, Tao Ni, Yongliang Chen, Di Duan, Kai Liu, and Weitao Xu. 2024. RF-Egg: An RF Solution for Fine-Grained Multi-Target and Multi-Task Egg Incubation Sensing. In *ACM MobiCom*.
[19] Shuai Tong, Jiliang Wang, and Yunhao Liu. 2020. Combating packet collisions using non-stationary signal scaling in LPWANs. In *ACM MobiSys*.
[20] Shuai Tong, Zhenqiang Xu, and Jiliang Wang. 2020. Colora: Enabling multi-packet reception in LoRa. In *IEEE INFOCOM*.
[21] Lei Wang, Tao Gu, Wei Li, Haipeng Dai, Yong Zhang, Dongxiao Yu, Chenren Xu, and Daqing Zhang. 2023. DF-Sense: Multi-user Acoustic Sensing for Heartbeat Monitoring with Dualforming. In *ACM MobiSys*.
[22] Xiong Wang, Linghe Kong, Liang He, and Guihai Chen. 2019. mLoRa: A Multi-Packet Reception Protocol in LoRa networks. In *IEEE ICNP*.
[23] Zhi Wang, Beihong Jin, Siheng Li, Fusang Zhang, and Wenbo Zhang. 2023. ECG-grained Cardiac Monitoring Using UWB Signals. *ACM IMWUT* (2023).
[24] Chenhao Wu, Xuan Huang, Jun Huang, and Guoliang Xing. 2023. Enabling Ubiquitous WiFi Sensing with Beamforming Reports. In *ACM SIGCOMM*.
[25] Xianjin Xia, Ningning Hou, Yuanqing Zheng, and Tao Gu. 2021. PCube: Scaling LoRa Concurrent Transmissions with Reception Diversities. In *ACM Mobicom*.
[26] Xianjin Xia, Yuanqing Zheng, and Tao Gu. 2019. FTrack: Parallel Decoding for LoRa Transmissions. In *ACM SenSys*.
[27] Binbin Xie, Minhao Cui, Deepak Ganesan, and Jie Xiong. 2023. Boosting the Long Range Sensing Potential of LoRa. In *ACM MobiSys*.
[28] Binbin Xie, Minhao Cui, Deepak Ganesan, and Jie Xiong. 2025. Making LoRa Sensing Coexist with Communication. In *ACM MobiCom*.
[29] Binbin Xie, Deepak Ganesan, and Jie Xiong. 2022. Embracing LoRa Sensing with Device Mobility. In *ACM SenSys*.
[30] Binbin Xie and Jie Xiong. 2020. Combating interference for long range LoRa sensing. In *ACM SenSys*.
[31] Binbin Xie, Yuqing Yin, and Jie Xiong. 2021. Pushing the Limits of Long Range Wireless Sensing with LoRa. *ACM IMWUT* (2021).
[32] Zhenqiang Xu, Pengjin Xie, and Jiliang Wang. 2021. Pyramid: Real-Time LoRa Collision Decoding with Peak Tracking. In *IEEE INFOCOM*.
[33] Fu Yu, Xiaolong Zheng, Liang Liu, and Huadong Ma. 2023. Enabling Concurrency for Non-orthogonal LoRa Channels. In *ACM Mobicom*.
[34] Fu Yu, Xiaolong Zheng, Yuhao Ma, Liang Liu, and Huadong Ma. 2024. Resolve Cross-Channel Interference for LoRa. In *IEEE ICDCS*.
[35] Gang Yu and Yiqi Zhou. 2016. General linear chirplet transform. *Mechanical Systems and Signal Processing* (2016).
[36] Fusang Zhang, Zhaoxin Chang, Kai Niu, Jie Xiong, Beihong Jin, Qin Lv, and Daqing Zhang. 2020. Exploring LoRa for Long-range Through-wall Sensing. *ACM IMWUT* (2020).
[37] Fusang Zhang, Zhaoxin Chang, Jie Xiong, Rong Zheng, Junqi Ma, Kai Niu, Beihong Jin, and Daqing Zhang. 2021. Unlocking the Beamforming Potential of LoRa for Long-range Multi-target Respiration Sensing. *ACM IMWUT* (2021).
[38] Jingang Zhong and Yu Huang. 2010. Time-Frequency Representation Based on an Adaptive Short-Time Fourier Transform. *IEEE Transactions on Signal Processing* (2010).